

Reaction Lens Systems: An Experience Report on Verifiable Full-Catalog Scientific Retrieval

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Abstract

Scientific retrieval software must preserve specified scoring behavior while its candidate collection and execution strategy change. We report an engineering evaluation of Reaction Lens Systems, an existing system that ranks Reactome reactions from English scientific text. We derive candidate-transformation contracts from its pointwise global/local scorer and check its blocked implementation against an unblocked reference. On sampled development inputs, 768 checks covering 10545024 score comparisons produce maximum observed logit difference 0.0 and 0 fixed-threshold decision changes. For a catalog of 15692 reactions, median cached-feature scoring time is 0.642 ms with the default block size, while median complete local prediction time is 268.90 ms on a separate convenience sample. An isolated installation also reproduces the checkpoint’s recorded development ranking metrics. The contribution is an artifact-backed account of execution contracts, component costs and reproducibility boundaries in one retrieval system; it is not a new learning algorithm or a claim of general retrieval superiority.

Code and data: <https://github.com/iCodeCraft/reaction-lens-systems>.

1 Problem and contribution

A researcher inspecting a scientific abstract may need to identify relevant biological reactions in a curated catalog. Reaction Lens Systems supplies ranked proposals for human review. It does not generate reactions or establish that a proposed association is biologically correct. Its engineering problem is maintaining an inspectable relationship between input text, catalog entries and predictions. Changing the order of candidates or the scoring block size should not silently change the score assigned to an unchanged article–reaction pair.

This report studies an existing implementation rather than introducing a new retrieval service. Its contributions are an explicit execution contract with metamorphic checks, a controlled comparison of blocked and unblocked exact scoring, and a package that separates regeneration of recorded tables from fresh model inference. These results are intended to help maintainers distinguish changes in ranking caused by new competitors from unintended changes in pairwise scores, and to identify which component dominates local request cost. We do not claim that candidate independence or blocked matrix operations are novel.

2 System and related work

Reactome supplies the biological reaction catalog [2]. SPECTER and SPECTER2 provide citation-informed scientific document representations [1, 4]. Reaction Lens Systems freezes its encoder and uses a learned global/local head. We evaluate a fixed checkpoint to study execution equivalence, component costs and inference reproducibility.

The processing path is English title/abstract, deterministic sentence spans, frozen text encoding, projected reaction vectors, exact blocked scoring, then ranked and threshold-selected output. Reaction

vectors are prepared once during initialization. The Python API, CLI and local HTTP service share the prediction implementation. The implementation rejects simultaneous calls on one instance; we evaluate local sequential calls, not multi-user throughput. Bundle loading verifies listed file hashes and shapes. These checks detect mismatches against a manifest; they do not authenticate a maliciously replaced manifest or guarantee semantic compatibility of arbitrary user-supplied encoders.

Metamorphic testing checks relations between transformed inputs and outputs when individual expected outputs are difficult to specify [3]. We instantiate this established approach for the candidate axis of the scorer. Agreement with an unblocked implementation checks execution equivalence, not biological correctness. Our reference shares the scoring implementation, so it cannot expose all common-mode defects. No fault-detection superiority is claimed.

3 Execution contracts

Let g and s_j be the normalized projections of the global article and valid sentence features through the shared learned article projection, and r the normalized output of the separate learned reaction projection. The fixed head computes

$$z(x, r) = a \left[g^\top r + w \max \left(0, \max_j s_j^\top r - n \right) \right] + b, \quad \hat{y}(x, r) = \mathbf{1}[z(x, r) \geq \tau]. \quad (1)$$

Here the learned scale a , local weight w , null n , bias b , and development-fitted threshold τ are held fixed. The implementation uses $a = \exp(\min(\ell, \log 100))$, $w = \text{softplus}(u)$ and $n = \tanh(v)$ for learned scalars ℓ, u, v . Padding is masked before the sentence maximum. No reduction is taken over competing reactions. Consequently, in real arithmetic, permuting candidates permutes outputs; removing candidates preserves retained scores; and appending candidates preserves previous scores. Fixed-threshold decisions inherit this property. Rankings and top- k membership do not: new candidates can outrank existing ones. Updating descriptions, model weights, preprocessing or the threshold also falls outside this contract.

For batch size B , sentence count S , candidate count R and projected width P , local similarities require $O(BSRP)$ arithmetic. Candidate blocks of width K bound the principal local-similarity intermediate to $O(BSK)$ instead of $O(BSR)$. The final score matrix still uses $O(BR)$ storage, and the prepared reaction bank uses $O(RP)$ storage. Blocking is an execution tradeoff, not a retrieval shortlist or a reduction in the number of scored candidates.

4 Evaluation protocol

The protocol is included as `PROTOCOL.md`. Experiments use the fixed structured-description checkpoint and its checksummed cached development pack. All 15692 catalog reactions are scored. We sample 64 publications without replacement using NumPy seed 20261005 and run CPU FP32 with two PyTorch threads. The environment is macOS 26.6.1 on arm64, Python 3.12.14, PyTorch 2.14.1 and NumPy 2.5.3. The sandbox did not expose the CPU brand; these results should not be treated as a portable hardware performance specification.

Each block configuration receives five warmup queries and 3 timed repetitions per sampled publication. Timings exclude encoding, bundle loading, initial reaction projection and result serialization. Configurations run sequentially in fixed order, so thermal and background-load effects remain possible. Sample p95 values are descriptive quantiles of repeated measurements, not confidence bounds.

On the first repetition we check candidate permutation, a half-catalog subset, appending 128 duplicate reaction vectors, and an unblocked reference. The predeclared absolute logit tolerance is 10^{-5} . Duplicate additions exercise candidate-axis execution; they do not evaluate new biological descriptions. Threshold checks use the existing release threshold without recalibration.

Memory is measured separately in a fresh process for each configuration using the same padded batch of 16 sampled publications, with maximum sentence count 14. We record process high-water resident

memory including imports and loaded feature arrays. These are not incremental tensor allocations, and each configuration has only one memory trial. Processes used for this comparison do not execute the unblocked reference or the candidate-transformation probes.

Complete prediction measurements use 5 existing released English comparison examples and three repetitions after one warmup. Their selection was for demonstration, so they are a convenience sample. Prediction includes text encoding, scoring, ranking and payload construction. JSON serialization is recorded separately; HTTP/network costs are excluded. Model setup is timed separately.

5 Results

5.1 Candidate and execution equivalence

The checks produce 10545024 scalar comparisons across 768 query/configuration/scenario combinations. The maximum observed absolute logit error is 0.0, with 0 fixed-threshold flips. These measurements support the intended contract on this checkpoint, platform and sample. They are not a universal bitwise-equivalence proof across backends. Evidence: [contract observations](#), derived from all individual probes in `results/system-benchmark.json`. Regenerate with `make contracts`.

5.2 Blocking and local execution cost

Block width	Median (ms)	p95 (ms)	Peak RSS (MiB)
128	1.676	1.827	412.53
1024	0.642	0.701	414.17
15692	0.656	0.764	444.05

Table 1: Cached-feature single-publication scoring latency and separate padded-batch process memory. These are different workloads. Every timing row scores the full catalog. Evidence: [block comparison](#); individual timings and memory observations are in `results/system-benchmark.json`. Regenerate with `make table1`.

The smallest block incurs greater observed latency. The default and unblocked configurations have similar medians; this sample does not establish a significant speed difference between them. The blocked configurations have lower observed process peak RSS in the separate batch probe, consistent with the intermediate storage bound. Memory differences should not be generalized from a single trial.

The 15 complete local predictions have median 268.90 ms and p95 393.54 ms. Encoding accounts for 96.33% of aggregate measured prediction time. Setup takes 4.87 s, and process peak RSS reaches 1149.22 MiB. Evidence: [component timings](#) and its raw source. Regenerate with `make timings`. For these examples, improving only head execution offers limited benefit to total latency because encoding dominates. This observation does not measure the effect of encoder caching, alternative encoders or approximate retrieval.

5.3 Reproduction boundary

An isolated dependency environment recomputes predictions from the downloaded checkpoint and cached features for 2377 development publications, 4094 known links and the full catalog. MAP is 0.4770, Hit@1 is 0.3925, and micro Recall@100 is 0.9016. Maximum absolute difference from the stored release metrics is 0.0. Evidence: [fresh evaluation](#). Recorded links are incomplete, and development data were used during model development. Recall@100 is not top-one accuracy or evidence of biological correctness.

For publication i , average precision sums the precision at each relevant rank over the full catalog and divides by the number of distinct recorded reaction labels for that publication, including labels absent from the catalog. MAP averages this value over publications. Hit@1 is the fraction of publications whose first result is recorded as relevant. Micro Recall@100 divides the total recorded links recovered

in the first 100 results by all recorded links across publications. Equal logits retain catalog order. These definitions and the acceptance tolerance 5×10^{-5} are implemented in [the evaluation script](#).

This exercise verifies packaging and cached-feature inference. It does not independently reproduce encoder training, head training or every historical experiment. The complete-text latency experiment exercises the local encoder but does not rebuild the entire development feature cache. The reserved test remains unopened. The measured scorer, encoder, pipeline and benchmark sources are preserved by checksums; packaging, validation and documentation can be revised separately.

6 Limitations and practical implications

This is a single-system, single-checkpoint, single-host experience report. It does not demonstrate a general improvement over vector databases, other retrieval architectures or other testing methods. Candidate independence follows directly from pointwise scoring; the evidence is about an implementation’s conformance and costs, not discovery of a new mathematical property. We have not measured failures in deployed systems, mutation-detection rates, user productivity or maintenance effort. The contract checks contain related scalar comparisons, not independent statistical samples. The convenience text sample and repeated development use limit semantic and performance generalization.

The reusable engineering lessons are to state whether a score is pairwise or candidate-set-dependent, distinguish score stability from rank stability, test blocking separately from retrieval quality, and report encoder and head costs separately. For reproduction, pin the model and feature artifacts and distinguish reconstruction of tables from actual inference. These practices can make this system easier to inspect, but our study does not quantify developer benefits.

7 Conclusion and availability

Reaction Lens Systems provides a concrete case in which exact catalog scoring can be blocked without observed changes to retained candidates’ scores, while encoder execution dominates measured local request latency. The artifact records all probes and ties displayed results to raw measurements through a generated `NUMBERS.csv`. The accompanying instructions provide table reconstruction, fresh inference, and experiment rerun commands as distinct operations. The code is licensed under MIT and the manuscript under CC BY 4.0.

Tool assistance. An AI coding assistant assisted with the measurement scripts, analysis and manuscript preparation. Reported measurements were produced by executing the scripts. The author is responsible for the content.

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